Nathan Hilton

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Work Experience

Pinterest **Jul 2022 – Present**

Software Engineer II (Mar 2024 – Present) - AI Builder Tools Team *San Francisco, CA, USA*

- Unlocked AI development with Codex and Claude for 2,000 monthly users by restricting privileged credentials and open internet access in a Docker image
- Further restricted those permissions while improving the developer experience by moving agents onto a new EC2 development environment protected by firewall rules & more fine-grained data access
- Scaling safe, sandboxed coding environments faster than existing environments by building on-demand, ephemeral Kubernetes development environments
- Reduced infrastructure costs by $1M annually by implementing lifecycle automation and approval workflows for GPU EC2 instances powering cloud development environments, enabling reliable start/stop and scheduled shutdowns
- Serve as GitHub platform lead, driving improvements to repository workflows and developer tooling across engineering

Software Engineer I (Jul 2022 – Feb 2024) - Code & Language Runtime Team

- Enabled adoption of GitHub for large multi-team monorepos by leading the migration of two complex monorepos from Phabricator, establishing the migration path for other teams
- Improved scalability of large monorepo workflows by implementing merge queues and standardized CI/CD pipelines
- Improved build reliability and developer workflows by building automation and tooling for monorepo development
- Expanded a Java/Bazel monorepo into a multi-language repository, introducing the first Python service and migrating a Go/Bazel monorepo to support Java, Python, and Go

American Express **May 2021 – Aug 2021**

Software Engineering Intern *New York, NY, USA*

- Built a production full-stack web application enabling automated keyword extraction from large text documents
- Designed a backend NLP pipeline and API used by internal teams to quickly analyze large documents, reducing manual review time

Space ML **Jan 2021 – Sep 2021**

Artificial Intelligence Researcher *Remote*

- Our project was selected as a Top 5 AI proposal for NASA SMD's Search for Groundbreaking Science
- Built a machine learning pipeline that runs a machine learning model efficiently and reliably on decades of satellite data
- Pipeline is automated, fully scalable, and very efficient, which has saved thousands of dollars

University of Florida **Jan 2021 – May 2022**

Teaching Assistant - Programming Fundamentals II *Gainesville, FL, USA*

- Led discussion sections of 25+ students, collaborating with TAs and professor to teach, grade, and host office hours
- Supported a 500+ student introductory CS class, providing guidance and feedback
- Strengthened communication skills and C++ expertise to help students grasp complex concepts

Projects

Battle of the Beats **Jan 2022 – May 2022**

Multiplayer Music Game *Gainesville, FL, USA*

- Built a browser-based multiplayer music game allowing users to create beats, melodies, and lyrics collaboratively
- Implemented peer-to-peer real-time communication to synchronize game state and support scalable multiplayer sessions
- Developed a modular digital audio workstation for composing and layering instruments

Technical Skills

Languages & Frameworks: Python, JavaScript/TypeScript, Go, C++, Node.js, GraphQL, Flask, React, Redux

AI & Developer Tools: Claude, Codex, VS Code, Cursor

DevOps & CI/CD: Git/GitHub, Bazel, Buildkite/Jenkins/GitHub Actions, Merge Queues, Docker, Kubernetes, AWS

Education

University of Florida - GPA: 3.83 **Jul 2018– May 2022**

Bachelor of Science in Computer Science, Minor in Business Administration *Gainesville, FL, USA*